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Change history

1.0.0	07/10/2026	First release
1.0.0	08/10/2026	Fix R1234 value due to issue xyz etc etc etc. Update C14 to 10 nF to reduce roll-off frequency.
1.0.0	TBD	TBD
1.0.0	TBD	TBD
1.0.0	TBD	TBD
1.0.0	TBD	TBD
1.0.0	TBD	TBD
1.0.0	TBD	TBD
1.0.0	TBD	TBD

ERC FAILURE

CHECK FAILED — SEE ERC REPORT

System overview

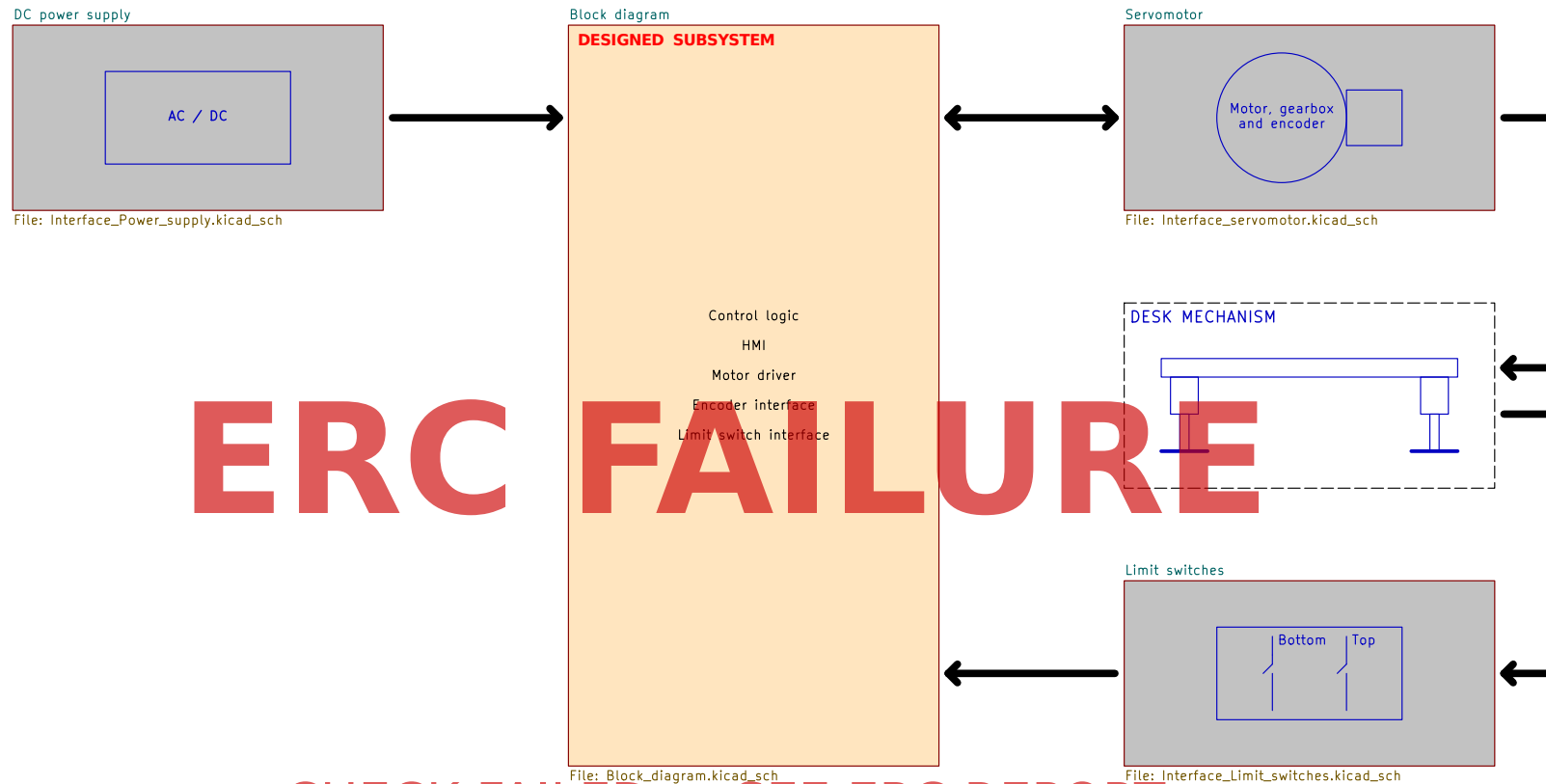


File: System_overview.kicad_sch



PROJNAME		Sheet: Root	
		File: ktest.kicad_sch	
Notes This is a comment 1 field - potentially not used This is a comment 2 field - potentially not used This is a comment 3 field - potentially not used This is a comment 4 field - potentially not used	Rev: 0.0.1	Drwn: FLR	
	Mat: Concept	Rev'd: FLR	
	Date: 2026-10-06	Rls'd: FLR	
	Dwg: NA	Size: A4	
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System overview



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This sheet shows the overall system and the interfaces to external components.
External components are shown for system context only and are not part of this schematic.
Only the "DESIGNED SUBSYSTEM" is covered by this schematic.

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PROJNAME

Notes

Sheet: System overview

File: System_overview.kicad_sch

Rev:

Mat: Concept

Date:

Dwg: NA

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Rev'd: FLR

RLs'd: FLR

Size: A4

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Interface specification

Barrel connector (Dell type, with ID pin)

Pin

- 1 Supply - 19.5 V, 3 A (inner shell)
- 2 ID (internal pin)
- 3 GND (outer shell)
- SH Shield

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PROJNAME

Notes

Sheet: DC power supply	
File: Interface_Power_supply.kicad_sch	
Rev:	Drwn: FLR
Mat: Concept	Rev'd: FLR
Date:	RLs'd: FLR
Dwg: NA	Size: A4
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Interface specification

JST B6B-XH-A / JST XHP-6 connector

Pin	Description
1	Encoder B output (VCC)
2	Encoder A output (VCC)
3	GND
4	VCC (3.3 - 5V)
5	Motor B (3 A, 12-24 V)
6	Motor A (3 A, 12-24 V)

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PROJNAME

Notes

Sheet: Servomotor

File: Interface_servomotor.kicad_sch

Rev:

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Date:

Dwg: NA

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Rev'd: FLR

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Interface specification

JST B4B XH A / JST XHP-4 connector

Pin Description

- 1 Bottom limit switch (Normally open, switch closes to GND)
- 2 GND
- 3 GND
- 4 Top limit switch (Normally open, switch closes to GND)

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PROJNAME

Notes

Sheet: Limit switches

File: Interface_Limit_switches.kicad_sch

Rev:

Mat: Concept

Date:

Dwg: NA

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Drwn: FLR

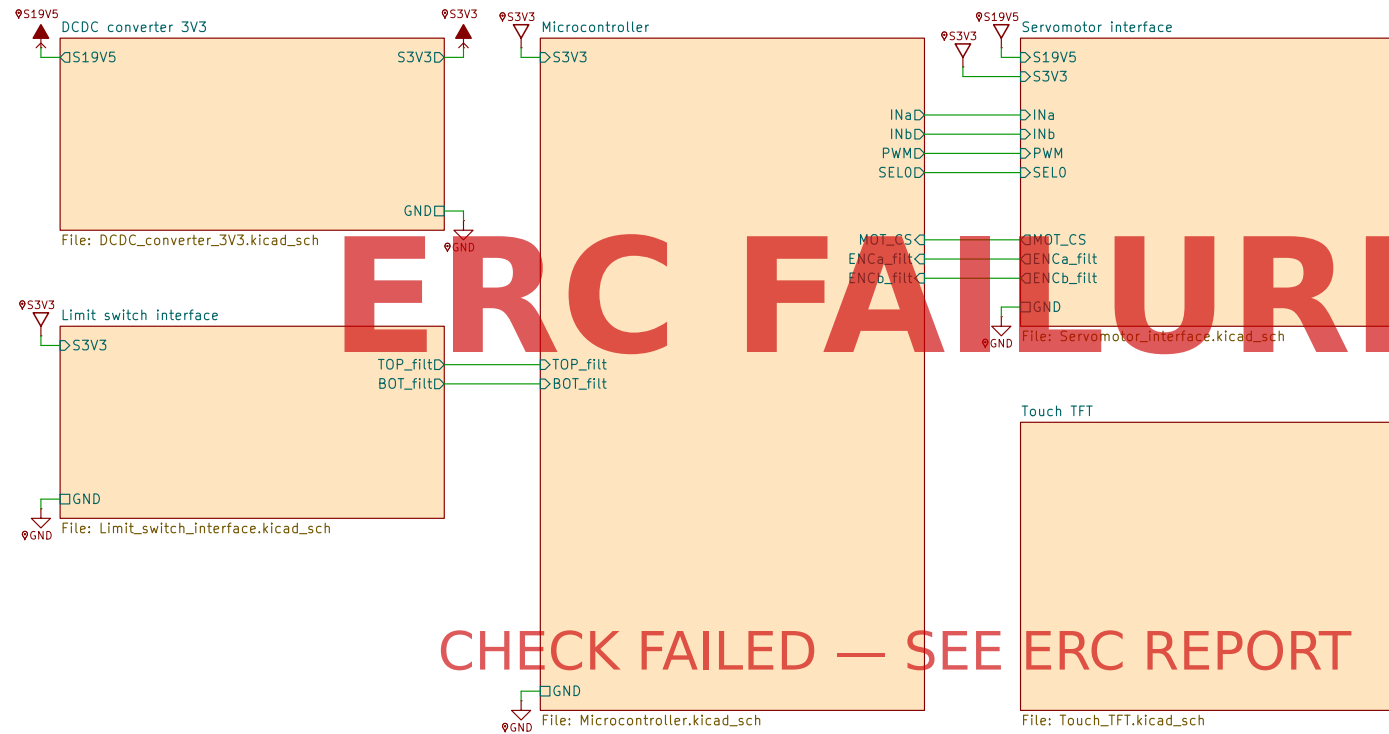
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Size: A4

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Block diagram



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PROJNAME

Notes

Sheet: Block diagram

File: Block_diagram.kicad_sch

Rev:

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Dwg: NA

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Rev'd: FLR

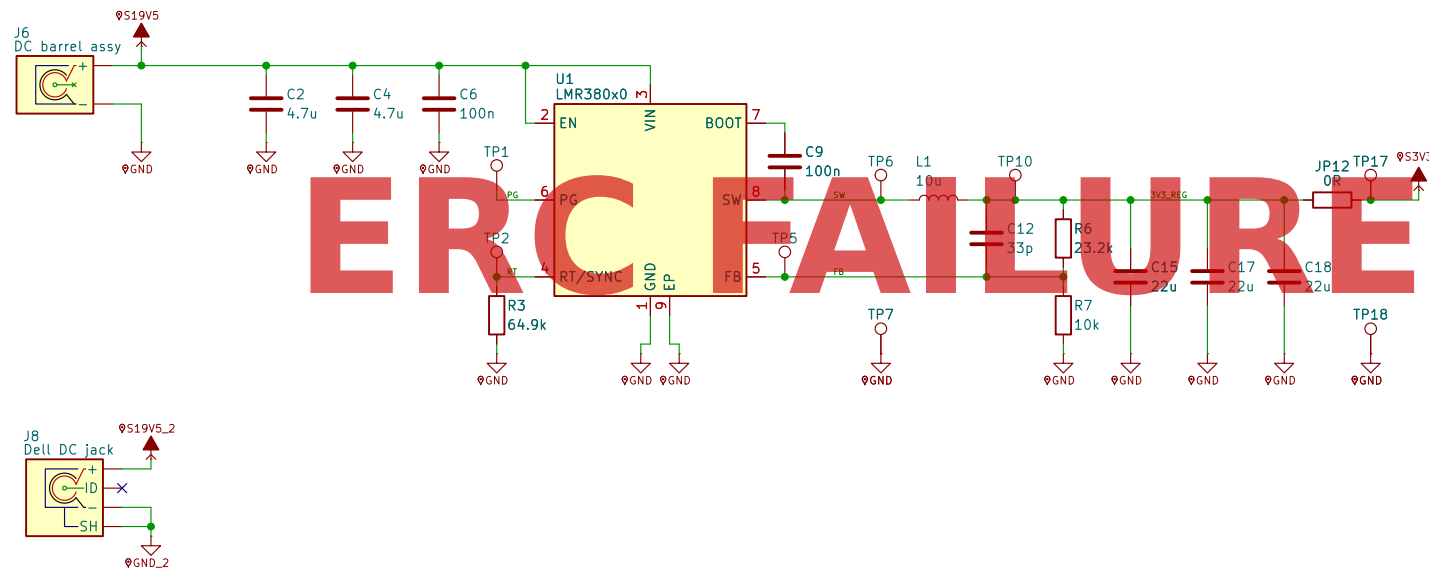
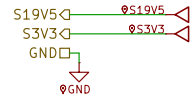
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DCDC converter 3V3

Block interface



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PROJNAME

Notes

Sheet: DCDC converter 3V3

File: DCDC_converter_3V3.kicad_sch

Rev:

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Date:

Dwg: NA

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Drwn: FLR

Rev'd: FLR

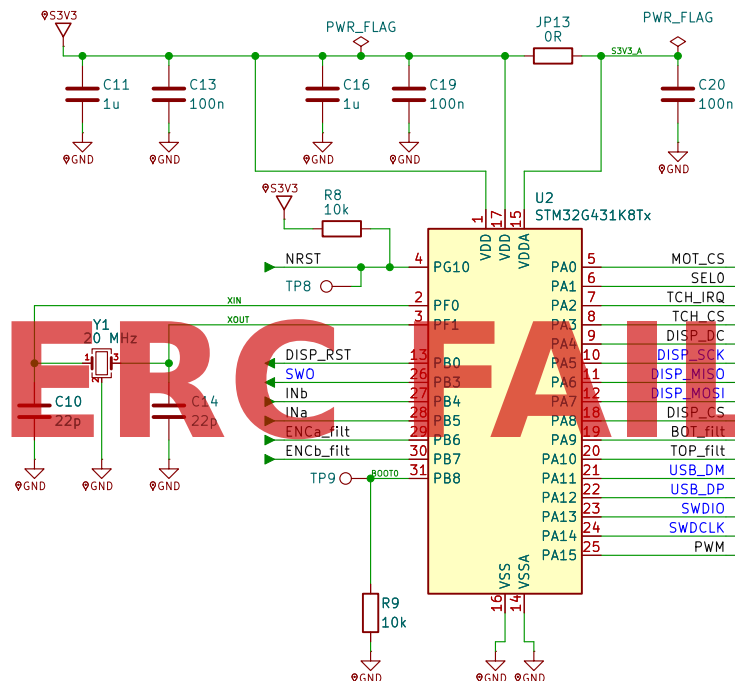
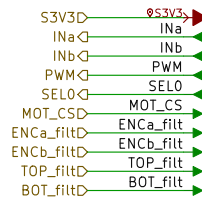
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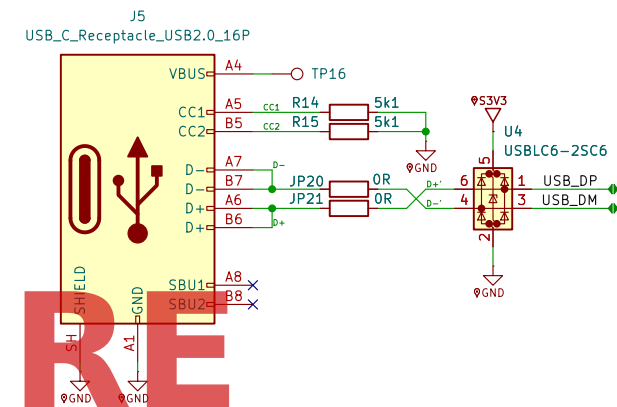
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Microcontroller

Block interface

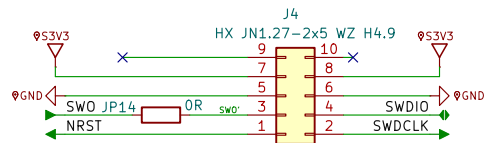


USB interface (serial comms port)



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SWD header



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PROJNAME

Notes

Sheet: Microcontroller

File: Microcontroller.kicad_sch

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Rev'd: FLR

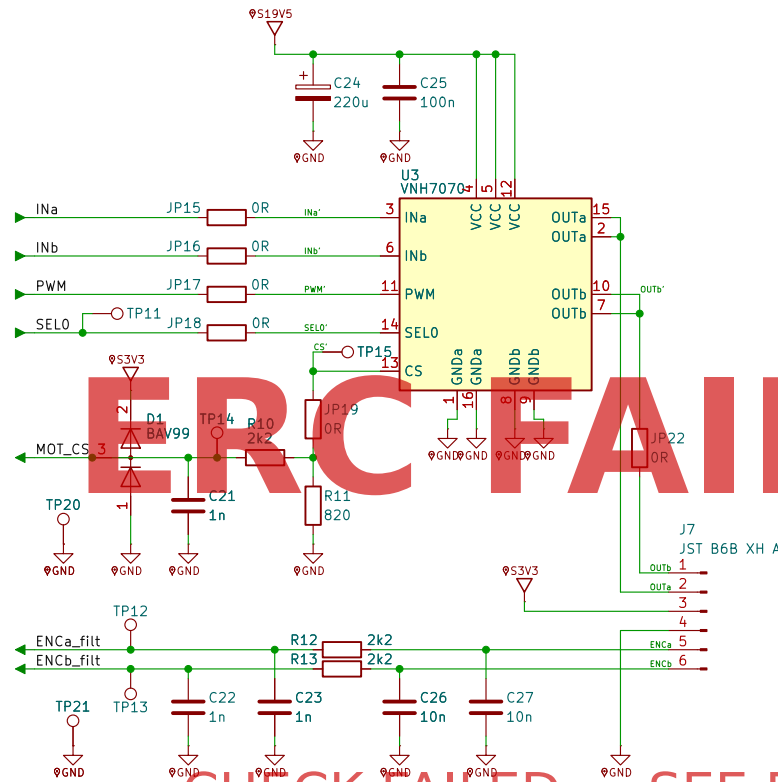
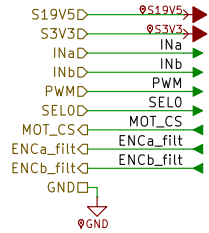
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Servomotor interface

Block interface



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PROJNAME

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Sheet: Servomotor interface

File: Servomotor_interface.kicad_sch

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Rev'd: FLR

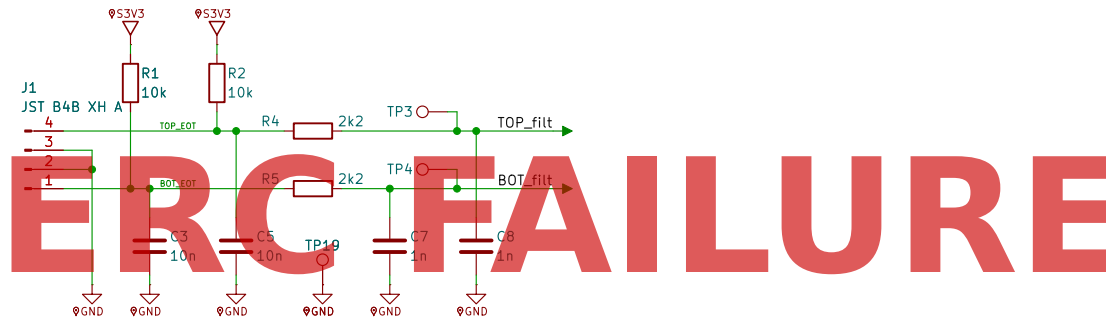
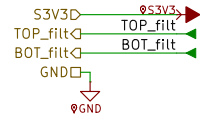
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Limit switch interface

Block interface



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PROJNAME

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